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1. (a) Let R be a finite commutative ring with unity. Prove that every nonzero element of R is either a zero-divisor or a unit.

Proof. Let R be a finite commutative ring with unity and let $a \in R$ such that $a \neq 0$. Since R is finite, the set $\{a^1, a^2, a^3, \dots\}$ must contain duplicates. So, there must be positive integers m, n with $m > n$ such that $a^m = a^n$. So,

$$\begin{aligned} a^m - a^n &= 0 \\ a^n(a^{m-n} - 1) &= 0 \end{aligned}$$

Case 1: Suppose $a^{m-n} - 1 = 0$. Then $a^{m-n} = 1$, so $a \cdot a^{m-n-1} = 1$. Thus, a is a unit with inverse a^{m-n-1} .

Case 2: Suppose $a^{m-n} - 1 \neq 0$. Then it must be that $a^n(a^{m-n} - 1) = 0$. Let $k \in \mathbb{Z}$ such that $a^k(a^{m-n} - 1) = 0$. If $k = 1$, then $a(a^{m-n} - 1) = 0$ which makes a a zero-divisor. If $k > 1$, then by factoring out an a from a^k ,

$$a \cdot [a^{k-1}(a^{m-n} - 1)] = a^k(a^{m-n} - 1) = 0$$

so a is still a zero-divisor. In either case, every nonzero element of R is a zero-divisor or a unit. $\square$

- (b) Give an example of a nonzero element in some ring that is neither a zero-divisor nor a unit.

Consider the ring $\mathbb{Z}$ and the element 2. There does not exist a nonzero element $b \in \mathbb{Z}$ such that $2b = 0$. 2 also can not be a unit since its multiplicative inverse, $\frac{1}{2}$, is not in $\mathbb{Z}$.

2. A ring element a is called an *idempotent* if $a^2 = a$.

- (a) Prove that the only idempotents in an integral domain are 0 and 1.

Proof. Let D be an integral domain and let $a \in D$ such that a is an idempotent. By contradiction, suppose $a \notin \{0, 1\}$. Then,

$$\begin{aligned} a^2 &= a \\ \Rightarrow a^2 - a &= 0 \end{aligned}$$

By Theorem 12.1,

$$\begin{aligned} a^2 - a &= a(a - 1) = 0 \\ \Rightarrow a &= 0 \text{ or } a - 1 = 0 \Rightarrow a = 1 \end{aligned}$$

This contradicts the assumption of $a \notin \{0, 1\}$. Thus, the only idempotents in D are 0 and 1. $\square$

- (b) Prove that if a is an idempotent in a ring, then $a^n = a$ for all positive integers n .

Proof. Let R be a ring and let $a \in R$ such that a is an idempotent. Since we want to show that any positive power of a is idempotent, we will proceed by induction.

Base Case: Let $n = 1$. Then, $a^1 = a$.

Inductive Step: Let $k \in \mathbb{Z}$ where $k > 1$ and suppose $a^k = a$. We want to show that the same is true for $k + 1$. So,

$$\begin{aligned} a^{k+1} &= a^k \cdot a \\ &= a \cdot a \\ &= a^2 \\ &= a. \end{aligned}$$

So, $a^{k+1} = a$. By induction, $a^n = a \forall$ positive integers n . $\square$

- (c) Find a nontrivial idempotent in $\mathbb{Z}_5[i] = \{a + bi \mid a, b \in \mathbb{Z}_5\}$.

The elements in $\mathbb{Z}_5[i]$ include

$$\begin{aligned} \mathbb{Z}_5[i] &= \{0, 1, 2, 3, 4, \quad (b = 0) \\ &\quad i, 1 + i, 2 + i, 3 + i, 4 + i, \quad (b = 1) \\ &\quad 2i, 1 + 2i, 2 + 2i, 3 + 2i, 4 + 2i, \quad (b = 2) \\ &\quad 3i, 1 + 3i, 2 + 3i, 3 + 3i, 4 + 3i, \quad (b = 3) \\ &\quad 4i, 1 + 4i, 2 + 4i, 3 + 4i, 4 + 4i\} \quad (b = 4) \end{aligned}$$

We can disregard the first row due to the modding by 5 making it impossible for the square of an element to get back to itself along with 0 and 1 being trivial idempotents. Similarly, the first column can be disregarded since taking the product i^2 along with the square of its coefficient will equal an integer in $\mathbb{Z}_5$, and in turn losing the i . Looking at the remaining elements, consider $3 + i$.

$$\begin{aligned} (3 + i)^2 &= (3 + i)(3 + i) \\ &= 9 + 6i - 1 \\ &= (8 \bmod 5) + (6 \bmod 5)i \\ &= 3 + i \end{aligned}$$

So, $\boxed{3 + i \text{ is an idempotent.}}$

3. (a) Let $S = \{a + bi \mid a, b \in \mathbb{Z}, b \text{ is even}\}$. Show that S is a subring of $\mathbb{Z}[i]$, but not an ideal of $\mathbb{Z}[i]$.

Proof. Let $S = \{a + bi \mid a, b \in \mathbb{Z}, b \text{ is even}\}$. By the Subring Test, we need to show that S is closed under subtraction and multiplication. Let $x_1 + x_2i, y_1 + y_2i \in S$ where $x_1, x_2, y_1, y_2 \in \mathbb{Z}$. Then,

$$\begin{aligned} x_1 + x_2i - (y_1 + y_2i) &= (x_1 - y_1) + (x_2 - y_2)i \\ (x_1 + x_2i)(y_1 + y_2i) &= x_1y_1 + (x_1y_2)i + (x_2y_1)i - (x_2y_2) \\ &= (x_1y_1 - x_2y_2) + (x_1y_2 + x_2y_1)i \end{aligned}$$

Since x_2 and y_2 are even, their difference must be even. Also, their products with x_1 and y_1 must also be even, making their sum of products even. So, $(x_1 + x_2i) - (y_1 + y_2i), (x_1 + x_2i)(y_1 + y_2i) \in S$ making S a subring of $\mathbb{Z}[i]$. Now we need to show that S is not an ideal of $\mathbb{Z}[i]$ by showing that $ab \notin S$ for some $a \in \mathbb{Z}[i]$ and $b \in S$. Consider $1 + i$ and $1 + 2i$. Then,

$$\begin{aligned} (1 + i)(1 + 2i) &= 1 + 2i + i - 2 \\ &= -1 + 3i \end{aligned}$$

Since 3 is odd, $-1 + 3i \notin S$. Thus, S is not an ideal of $\mathbb{Z}[i]$. □

- (b) Let

$$R = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in \mathbb{R} \right\} \quad \text{and} \quad A = \left\{ \begin{bmatrix} 0 & x \\ 0 & y \end{bmatrix} \mid x, y \in \mathbb{R} \right\}$$

Prove that A is an ideal of R , and that R/A is a field.

Proof. Let R and A be as defined in the problem statement. To show that A is an ideal of R , we need to show that ab and $ba \in A$ for all $a \in A$ and $b \in R$. Let $a = \begin{bmatrix} a_1 & b_1 \\ 0 & c_1 \end{bmatrix}$ and $b = \begin{bmatrix} 0 & x_1 \\ 0 & y_1 \end{bmatrix}$ where $a_1, b_1, c_1, x_1, y_1 \in \mathbb{R}$. Then,

$$\begin{aligned} ab &= \begin{bmatrix} a_1 & b_1 \\ 0 & c_1 \end{bmatrix} \begin{bmatrix} 0 & x_1 \\ 0 & y_1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & a_1x_1 + b_1y_1 \\ 0 & c_1y_1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} ba &= \begin{bmatrix} 0 & x_1 \\ 0 & y_1 \end{bmatrix} \begin{bmatrix} a_1 & b_1 \\ 0 & c_1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & x_1c_1 \\ 0 & y_1c_1 \end{bmatrix} \end{aligned}$$

Since every non-zero entry is a real number, their products and sums are also real numbers. A is also closed under subtraction since the difference of two real numbers is also a real number and the left column will stay unchanged. Thus, A is an ideal of R .

By Theorem 14.4, R/A is a field iff A is maximal.

□

4. (a) If an ideal I of a ring R contains a unit, show that $I = R$.

Proof. Let I be an ideal of a ring R and let $a \in I$ be a unit. By definition of a unit, there exists an $a^{-1} \in R$. Since I is an ideal, $aa^{-1} \in I$. Thus, $1 \in I$. Since $1 \in I$, $r \cdot 1 = r \in I$ for all $r \in R$. That means $R \subseteq I$. Thus, $I = R$. □

- (b) Let R be a commutative ring with unity. Prove that R is a field if and only if the only ideals of R are $\{0\}$ and R .

Proof. ($\Rightarrow$) Suppose R is a field. Let I be an ideal of R . Then I contains some nonzero element a . Since R is a field, a is a unit. By part (a), $I = R$.

($\Leftarrow$) Suppose the only ideals of R are $\{0\}$ and R . Let $a \in R$ with $a \neq 0$. Consider the principal ideal $\langle a \rangle = \{ra \mid r \in R\}$. Since $a \neq 0$, $\langle a \rangle \neq \{0\}$, so the only option is for $\langle a \rangle = R$. In particular, $1 \in \langle a \rangle$, so there exists $r \in R$ with $ra = 1$. Thus a is a unit. Since every nonzero element of R is a unit, R is a field. □

5. (a) Prove that $I = \langle 2 + 2i \rangle$ is not a prime ideal of $\mathbb{Z}[i]$.

Proof. We need to find elements $a, b \in \mathbb{Z}[i]$ such that $ab \in \langle 2 + 2i \rangle$ but $a \notin \langle 2 + 2i \rangle$ and $b \notin \langle 2 + 2i \rangle$. Consider $a = b = 2$. Then $ab = 4$, and

$$(1 - i)(2 + 2i) = 2 + 2i - 2i - 2i^2 = 2 + 2 = 4$$

so $4 \in \langle 2 + 2i \rangle$. Now suppose $2 \in \langle 2 + 2i \rangle$. Then there exists $x + yi \in \mathbb{Z}[i]$ such that $(x + yi)(2 + 2i) = 2$. Expanding,

$$(2x - 2y) + (2x + 2y)i = 2 + 0i$$

So $2x - 2y = 2$ and $2x + 2y = 0$. Adding these gives $4x = 2$, so $x = \frac{1}{2} \notin \mathbb{Z}$, a contradiction. Thus $2 \notin \langle 2 + 2i \rangle$. Since $2 \cdot 2 = 4 \in \langle 2 + 2i \rangle$ but $2 \notin \langle 2 + 2i \rangle$, $\langle 2 + 2i \rangle$ is not a prime ideal. □

- (b) Let $n = st$, where s and t are divisors of n greater than 1. Prove that $\langle s \rangle$ is a maximal ideal in $\mathbb{Z}_n$ if and only if s is prime.

Proof. ($\Rightarrow$) Suppose $\langle s \rangle$ is a maximal ideal of $\mathbb{Z}_n$. By contradiction, suppose s is not prime. Then $s = pq$ for some positive integers p, q with $p < s$. Since p divides s , multiples of s are also multiples of p . But since $p < s$, p cannot be a multiple of s in $\mathbb{Z}_n$, thus $p \notin \langle s \rangle$ and $\langle s \rangle \subsetneq \langle p \rangle$.

Furthermore, since $s \mid n$ and $p \mid s$, then $p \mid n$. Because $p > 1$, $\gcd(p, n) \geq p > 1$, which implies $1 \notin \langle p \rangle$. Thus, $\langle p \rangle \neq \mathbb{Z}_n$. This gives us $\langle s \rangle \subsetneq \langle p \rangle \subsetneq \mathbb{Z}_n$, which contradicts $\langle s \rangle$ being maximal. Thus, s must be prime.

($\Leftarrow$) Suppose s is prime. Let B be an ideal of $\mathbb{Z}_n$ such that $\langle s \rangle \subseteq B$ and $B \neq \langle s \rangle$. We want to show $B = \mathbb{Z}_n$. Since $B \neq \langle s \rangle$, there exists $b \in B$ such that $b \notin \langle s \rangle$. In regards to s , this means b is not a multiple of s , so $s \nmid b$. Since s is prime and $s \nmid b$, that means the $\gcd(s, b) = 1$. By Bézout's Identity, there exist integers u, v such that $us + vb = 1$. Since $s \in \langle s \rangle \subseteq B$ and $b \in B$, and ideals are closed under multiplication and subtraction, which is the inverse of addition, so $us + vb$ must be in B . So, $1 \in B$. Thus, $B = \mathbb{Z}_n$. Thus, $\langle s \rangle$ is maximal. $\square$

6. If R is a finite commutative ring with unity, prove that every prime ideal of R is a maximal ideal of R .

Proof. Let P be a prime ideal of R . Then R/P is an integral domain. Since R is finite, R/P is also finite. By 1(a), every nonzero element of R/P is either a zero-divisor or a unit. Since R/P is an integral domain, it has no zero-divisors. Therefore every nonzero element of R/P is a unit, making R/P a field. Since R/P is a field, P is a maximal ideal. $\square$